

Decision Making

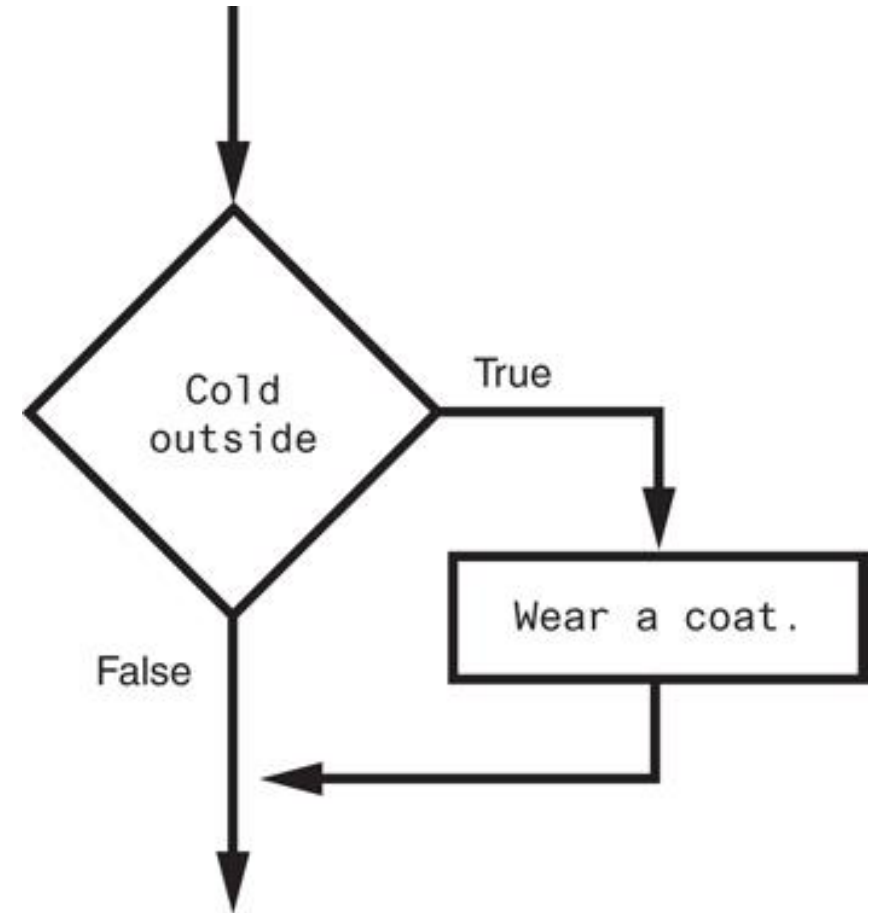
CSCI 1101 Introduction to
Computer Science

Topics

- ❖ if Statements
- ❖ if-else Statements
- ❖ if-elif-else Statements
- ❖ Nested Statements

if Statement Flowchart

- **If statements** are used in programming to make decisions
- In flowchart, diamond represents true/false condition that must be tested
- Actions can be *conditionally executed*
 - Performed only when a condition is true
- Single alternative decision structure: provides only one alternative path of execution
 - If condition is not true, exit the structure



if Statement Syntax

Python syntax:

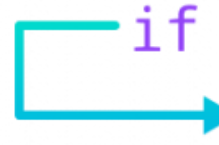
```
if condition:  
    statement  
    statement
```

- First line known as the `if` clause
 - Includes the keyword `if` followed by condition
 - The condition can be true or false
 - When the `if` statement executes, the condition is tested, and if it is true the block statements are executed. otherwise, block statements are skipped

if Statement Example

Condition is True

```
number = 10  
if number > 0:  
    # code  
  
# code after if
```



Condition is False

```
number = -5  
if number > 0:  
    # code  
  
# code after if
```



The **if-else** Statement

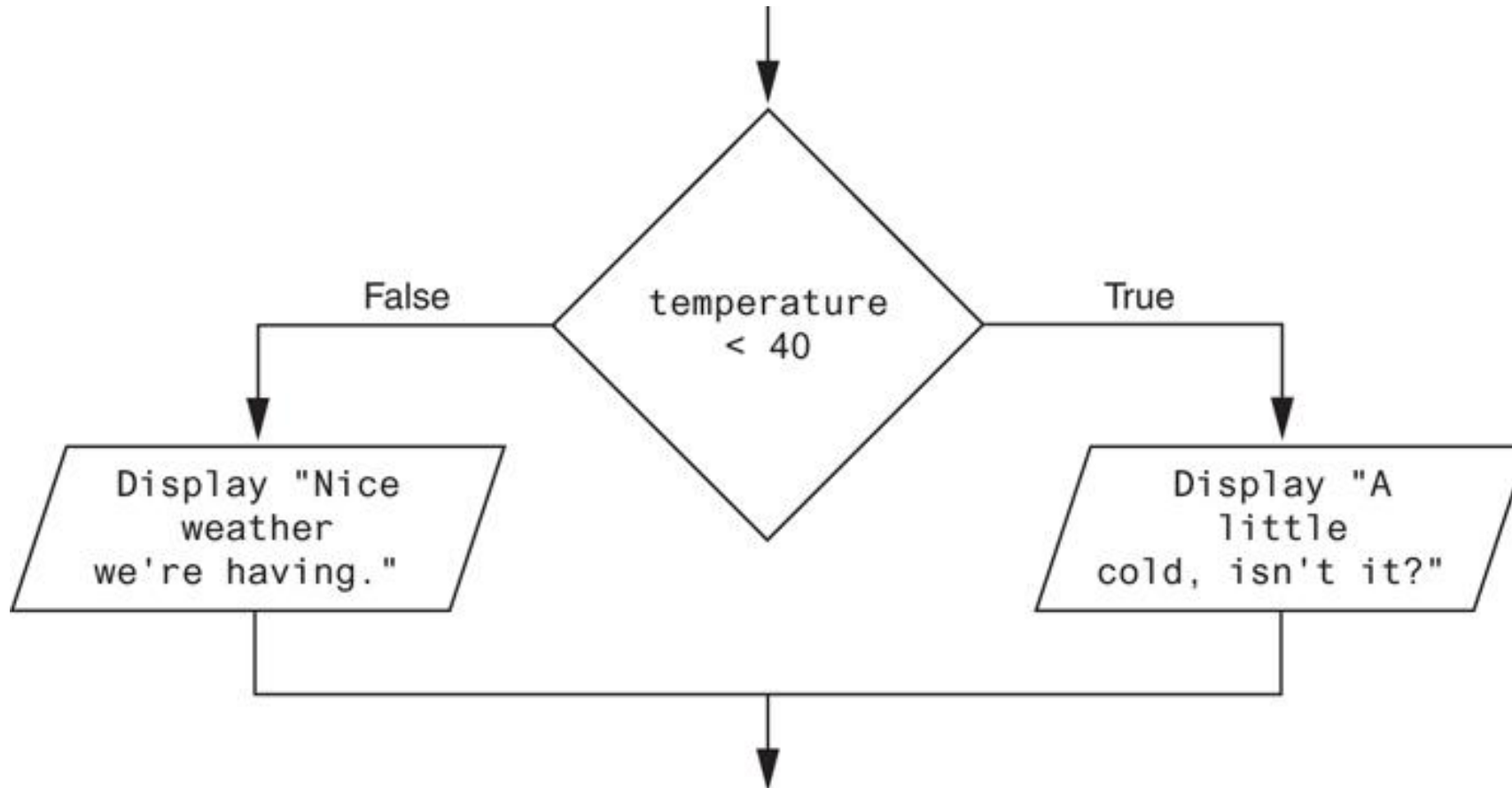
- Dual alternative decision structure: two possible paths of execution
One is taken if the condition is true, and the other if the condition is false

- **Syntax:**

```
if condition:
    statements
else:
    other statements
```

- `if` clause and `else` clause must be aligned
- Statements must be consistently indented

if-else Statement Workflow



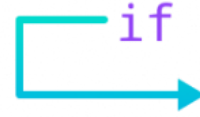
if-else Statement Example

Condition is True

```
number = 10
if number > 0:
    # code

else:
    # code

# code after if
```

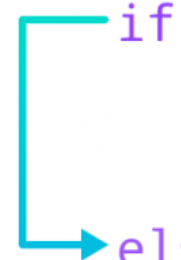


Condition is False

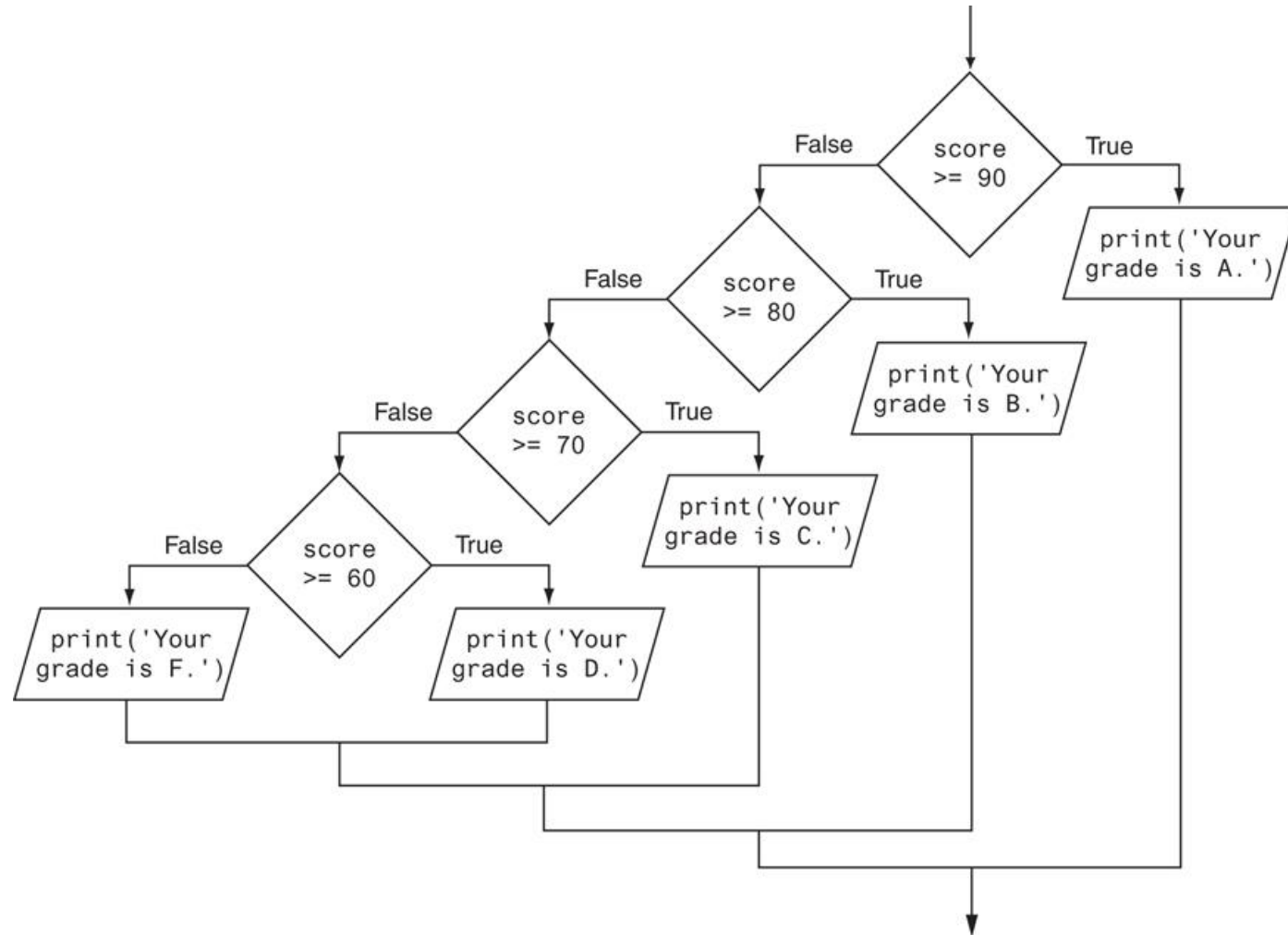
```
number = -5
if number > 0:
    # code

else:
    # code

# code after if
```



if-elif-else Statement Workflow



if-elif-else Statement Example

*Ignore the **let** keyword, it's not Python syntax.

1st Condition is True

```
let number = 5
if number > 0 :
    # code
elif number < 0 :
    # code
else :
    # code
# code after if
```

2nd Condition is True

```
let number = -5
if number > 0 :
    # code
elif number < 0 :
    # code
else :
    # code
# code after if
```

All Conditions are False

```
let number = 0
if number > 0 :
    # code
elif number < 0 :
    # code
else :
    # code
# code after if
```

if-elif-else Statements Rules

Important to use proper indentation in a nested decision structure

- Important for Python interpreter
- Makes code more readable for programmer
- Rules for writing nested if statements:
 - `else` clause should align with matching `if` clause
 - Statements in each block must be consistently indented

if-elif-else Statements Rules

if-elif-else statement: special version of a decision structure

- Makes logic of nested decision structures simpler to write
 - Can include multiple `elif` statements
- Syntax:

```
if condition_1:  
    statement(s)  
elif condition_2:  
    statement(s)  
elif condition_3:  
    statement(s)  
else  
    statement(s)
```

Insert as many `elif` clauses
as necessary.

Nested Statements

Inside of an `if-else` statement we write the code we want executed depending on the outcome of the conditional expression.

The code inside can be another `if-else` statement.

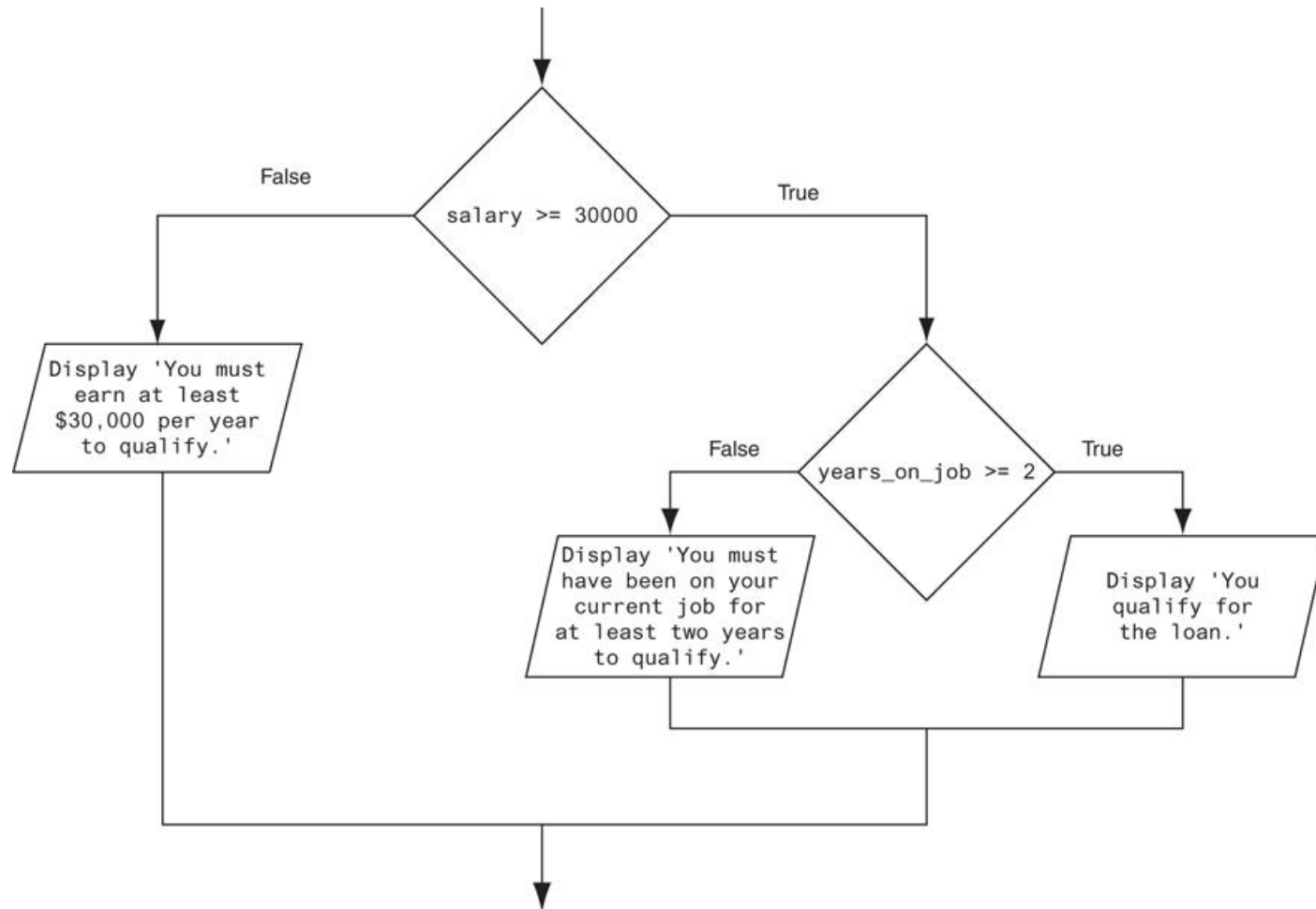
When doing this, make sure to double check your indentations.

The **if-else** Nested Statement

A decision structure can be nested inside another decision structure

- Commonly needed in programs
- **Example:**
 - Determine if someone qualifies for a loan, they must meet two conditions:
 - Must earn at least \$30,000/year
 - Must have been employed for at least two years
 - Check first condition, and if it is true, check second condition

Nested **if-else** Statement Workflow



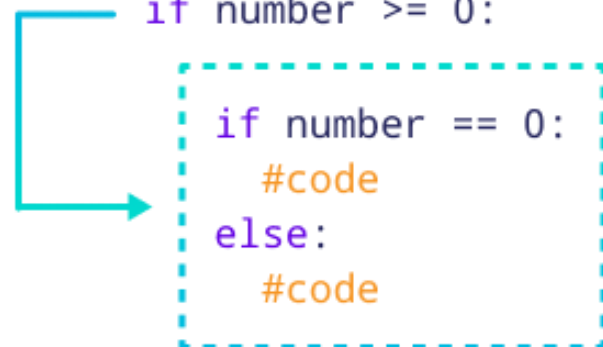
Nested if-else Statement Workflow

```
if salary >= 30000:
    if years_on_job >= 2:
        print("You qualify for the loan.")
    else:
        print("You must have been on your current job for at least two years to qualify.")
else:
    print("You must earn at least $30,000 per year to qualify.")
```


Nested Statements Example

Outer if Condition is True

```
number = 5
if number >= 0:
    if number == 0:
        #code
    else:
        #code
else:
    #code
```



Outer if Condition is False

```
number = -5
if number >= 0:
    if number == 0:
        #code
    else:
        #code
else:
    #code
```

